159102 workshop – Week 4

Random numbers

1. If a random number generator is producing numbers in the interval [55, 62) can it ever produce the number 62? Explain your answer.

2. Write a program to simulate throwing one dice 6000 times. Display the count for every number that can occur when throwing one dice.

3. Write a program that generates random numbers in the interval [2, 12]. Generate 6000 numbers and display the results for every number that can occur in the interval.

4. When throwing two dice, the result can be any number from 2 to 12. Some people think that every result has the same chance of occurring, but this is not correct. Write a program that simulates throwing two dice 6000 times and display all results. Discuss how it differs from the results in (3) above.
(Hint: throwing two dice is the sum of one dice + one dice).

5. Assume someone has invited you to throw two dice (only once). Before you throw the dice you must predict what the result will be. What number should you select as your prediction?

6. Thinking about computer simulations, have a look at Boids:
https://www.youtube.com/watch?v=mNZq0RhM-98
If you don't like the video, just google Boids and you will find lots of other videos. Boids doesn't really use random numbers (which actually makes it more amazing). This has nothing to do with our course but every Computer Scientist should know about Boids!

Machine Code

7. Write the correct MASSEY machine code instructions to carry out the following:
a) place the hexadecimal value 23 in register 0 and also in a memory location
b) OR the contents of registers 8 and 9 and store the result in a memory location
c) add the contents of two memory locations
d) jump to the instruction at memory address 13 if register 5 contains a 6

8. What value is in the program counter immediately after executing the instruction D0BD?

9. Divide the instructions below into two groups as follows:
Group A – instructions that involve memory location 2A
Group B – instructions that have nothing to do with memory location 2A

112A 202A 332A 442A 552A 662A 882A DD2A